

Day 2 — The Setup: Deficits from 10 (Single-Digit $\times$ Single-Digit)

Module: Nikhilam Multiplication · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will instantly state the deficit of any single digit from 10, AND will solve any single-digit $\times$ single-digit multiplication where both digits are 6–9 using the Nikhilam method.

Materials

- Class wall-poster: “Deficits from 10” — the table $1 \rightarrow 9, 2 \rightarrow 8, \dots, 9 \rightarrow 1$. Display before class.
- Drill sheet: 20 single-digit $\times$ single-digit problems (both factors 6–9).
- Stopwatch per pair.

Lesson flow

Warm-up (5 min)

- **Complement-to-10 lightning round.** Teacher calls 10 single digits in 30 seconds; class chants the deficit. ($7 \rightarrow 3, 8 \rightarrow 2, 6 \rightarrow 4, 9 \rightarrow 1, \dots$)
- Repeat with a different sequence. By the end most students should be calling out the deficit faster than they can think it.

Core (15 min) — Nikhilam at the single-digit level

1. **Set up the template on the board** (use the vertical line):

$$\begin{array}{r|l} 7 & -3 \\ \times 8 & -2 \\ \hline ? & ? \end{array} \quad \begin{array}{l} \leftarrow 7 \text{ is } 3 \text{ below } 10 \\ \leftarrow 8 \text{ is } 2 \text{ below } 10 \end{array}$$

2. **The two operations:**

- LEFT part: cross-subtract. $7 - 2 = 5$. (Or $8 - 3 = 5$. Either cross gives the same number — that’s a built-in check.)
- RIGHT part: multiply the deficits. $3 \times 2 = 6$.
- Concatenate: **56**.

3. **Verify.** $7 \times 8 = 56$. ✓

4. **Why does either cross give the same answer?** Quick proof on board: $7 - 2 = 7 - (10 - 8) = 7 + 8 - 10$ $8 - 3 = 8 - (10 - 7) = 7 + 8 - 10$ Same thing. Don’t dwell — Day 7 has the full algebra.

5. **The hand-trick (optional, fun).** Hold up both hands. For 7×8 : fold down 2 fingers on the left hand (deficit of 7 from 10 = 3, so keep 7 fingers up; or fold down $7 - 5 = 2$ on the left) and 3 on the right (deficit of $8 - 5 = 3$, so fold down $2 + 3 = 5$... use whichever finger-mnemonic the class likes). The number of fingers DOWN $\times 10$ = the tens; the number of fingers UP on each hand multiplied = the units. (This is a folk technique that maps to the Nikhilam algebra.) Show once, don't drill it.
6. **Worked examples in chorus:**
 - 9×8 : deficits 1, 2. Left: $9 - 2 = 7$. Right: $1 \times 2 = 2$. Answer: **72**. ✓
 - 6×9 : deficits 4, 1. Left: $6 - 1 = 5$. Right: $4 \times 1 = 4$. Answer: **54**. ✓
 - 6×7 : deficits 4, 3. Left: $6 - 3 = 3$. Right: $4 \times 3 = 12$. Hmm — right part is two digits! Carry the 1 to the left: left = $3 + 1 = 4$; right = 2. Answer: **42**. ✓
7. **Note the carry rule.** When the right part has more digits than the base has zeros (here: base 10, so right part should be 1 digit; if it's 2 digits, carry the extra). Write on board:

“If the right part overflows the base, carry the overflow into the left part.”

Activity (20 min) — Drill

Pair drills.

- Hand out the drill sheet — 20 problems, all single-digit $\times$ single-digit, both factors in 6–9.
- Working in pairs:
 - Partner A reads problem 1 aloud.
 - Partner B answers in under 3 seconds using Nikhilam.
 - Swap.
- 10 min of paired drill. Then 5 min of individual silent practice.

After 5 minutes of paired drills, do a class-level **timing test**: project 10 problems on the board; students answer all 10 individually in 60 seconds.

Wrap-up (5 min)

- Quick check: chorus answers — 7×7 ? (49.) 6×8 ? (48.) 9×9 ? (81.) 8×6 ? (48.)
- Preview Day 3: “Tomorrow — both numbers near 100. Same trick, slightly bigger.”

Student-facing content

Single-digit Nikhilam (base = 10)

1. Find each digit's **deficit from 10**.
2. Cross-subtract for the LEFT part.
3. Multiply the deficits for the RIGHT part.
4. Stick them together. If the RIGHT part is 2 digits, **carry** the overflow.

Examples: - 7×8 : $7-2=5$; $3 \times 2=6 \rightarrow 56$. - 9×8 : $9-2=7$; $1 \times 2=2 \rightarrow 72$. - 6×7 : $6-3=3$; $4 \times 3=12 \rightarrow$ carry 1 $\rightarrow 3+1=4$; right=2 $\rightarrow 42$.

The Sanskrit sūtra: **Nikhilam Navataścaramaṁ Daśataḥ** — “all from 9 and the last from 10.”

Homework

- Practice deficit-from-10 recall. Have a parent/sibling call out single digits 1–9; you answer the deficit. Beat 20 problems in 30 seconds.
- Solve 10 problems on the back of today’s worksheet (all single-digit $\times$ single-digit, both 6–9) using Nikhilam. Self-check against your times table.

Differentiation

- **Tier up:** Try $9 \times 9 \times 9$ (chained) or apply the method to base 20 — see if 19×18 works the same way with base 20. (Yes — deficits 1, 2 $\rightarrow$ left 17, right 2 $\rightarrow 17 \times 20 + 2 = 342$. ✓)
- **Tier down:** Stick to 8×9 , 9×9 , 7×8 for today. Memorise these specific three; we’ll widen the range tomorrow.

Teacher notes

- Some students will protest: “*But I already know $7 \times 8 = 56$ from my times tables!*” Honest answer: “*Yes — for single digits the times-table is faster. The Nikhilam method is here so we can use it on 2-digit numbers tomorrow, where the times-table doesn’t help.*”
- Watch for the carry rule on 6×7 and 6×8 . This is the most common error point.
- The “either cross gives the same” property is a built-in check students can use to catch errors. Reinforce it.

Citation(s) used in this lesson

- *Vedic Mathematics Batch1.pdf* (Vedic Cultural Center, Sammamish WA) — Nikhilam multiplication, single-digit examples.
- *Microsoft Word - sutras1.pdf* — sūtra #2: **Nikhilam Navataścaramaṁ Daśataḥ**.